# Dissecting Capital Flows: Do Capital Controls Shield Against Foreign Shocks?

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[1] The views in this paper are those of the authors and do not represent the OECD or of its members.

# Background

## A whole new world

- Changed economic environment: More financial integration, less (degree of) Original Sin
- EMEs can now invest more broadly (gross outflows from EMEs matter, Kalemli-Ozcan, 2019)

## New Dynamics of Capital Flows:

- Global Financial Crisis: Capital Flows $\downarrow$ to all locations (**global retrenchment** of 2008)
- Increased correlation of (gross) capital inflows and outflows (Forbes and Warnock, 2012; Davis and van Wincoop, 2021)

## Implications:

- Disentangling (net) flows into gross out and inflows becomes relevant
- Control measures are typically ill advised (e.g., Andreanssen, et al, 2023; Forbes, 2007, JIE)
- Yet, they are still actively applied. Why? … are there overseen benefits to them?

  Policy institutions are updating their outlook on CFMs (e.g., ASSA 2023, IMF Integrated Policy Framework '20)

# What we do

We analyze the insulation properties of Capital Flow Management measures (CFM).

Focus on disaggregating each relevant dimension:

- Direction: Net vs. Gross flows (total and separately —Out, In)
- Type of Policy: Any control, On Inflows only, On Outflows
- Type of Investment Flow: Total, Risky, Safe (and subcategories)
- Policy Regime: Tight and Loose monetary conditions

**Quantity of interest:** Mitigation of change in capital flows after a US monetary policy shock

- $\Rightarrow$ (research question) Can we shield —the capital flows of— a financially integrated economy with controls?

# Results overview:

- CFM measures can insulate emerging economies from foreign shocks.
- But the level of disaggregation is critical for assessing CFMs (of flows, assets, policies).
- Literature mitigation effect on Net flows is ambiguous (Magud, Reinhart, Rogoff, 2018; Reinhart and Smith, 2012; Erten, Korinek and Ocampo, 2021)

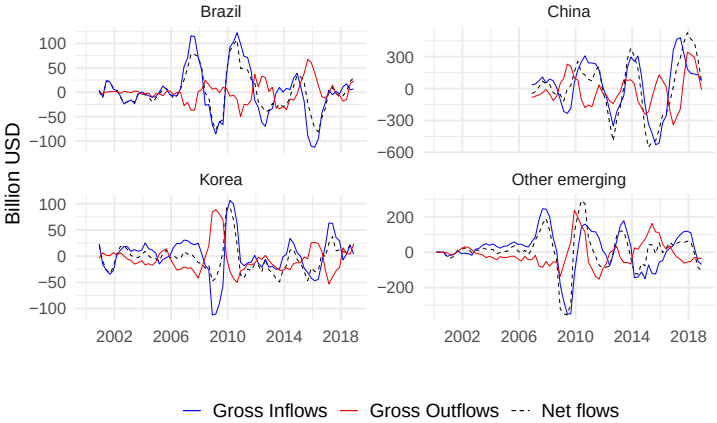
  Our take: Heterogeneity of effects across flows make evaluation —of CFMs— fuzzy. (e.g., effects may offset betwen types of gross flows.)
- Not evaluating at appropriately disaggregated level leads to (positive and negative) biases.
- Not all flows are in need of insulation: Only Risky investments and Banking flows.
- Insulation is weaker during tighter monetary conditions.

  $\Rightarrow$ **CFMs implementation and design should account for these nuances —to decrease any unnecessary interventionism.**

# Motivation

Nature of capital flows and financial markets has changed dramatically since the GFC of 2008

There's a ***stronger correlation between capital inflows and outflows***




Source: IMF - International Finance Statistics.

Capital outflows have gained traction in EMEs

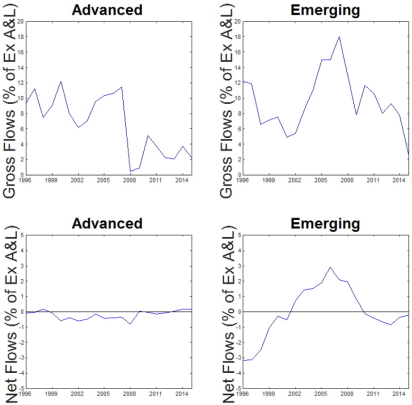
Inflows $\equiv$ Investment at home by foreign agents. (Can increase or decrease without affecting outflows).

Outflows $\equiv$ Investment abroad by home agents.

# Motivation (cont.)

**Before:** analysis concerned only Net Flows

With higher correlation of the gross outflows it's relevant to analyze each flow separately (Kalemli-Ozcan, 2019; Davis and Van Wincoop, 2021)

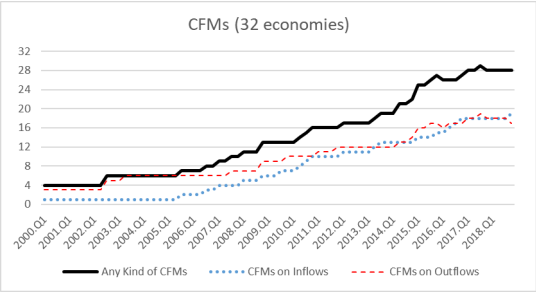



Source: Davis and Van Wincoop (2021, JIE).

Key: Changed extent of EMEs investments and debt issuance (original sin).

# Policy Responses

Despite the higher financial integration we have seen an increase in the application of CFMs.




Source: IMF

Reflection of new dynamics of capital flows and GFC experience

growth of CFM measures in outflow is not prevalent in pre-crisis years

Now it's even on par with measures on inflows

Literature reflects the changes: traditional focus $\rightarrow \Delta$ current account (net inflows/outflows)

[Now] Growing interest in analyzing Gross flows separately: Broner et al (2013), Bruno and Shin (2015), Cerutti, Claessens and Rose (2017), Davis and Van Wincoop (2018, 2021, 2023), among others.

# The Effect of CFM measures

CFMs, financial integration, and new flows' dynamics can have distinct economic effects

New flows' dynamics: Sign of higher financial integration $\rightarrow$ why to couple this tight controls?

**Traditional view:** CFMs limit integration, risk hedging, and have adverse effects on growth and on misallocation of exports (Andreanssen, Bauducco, Dardati, Mendoza, 2023).

At the applied extent CFMs may bring unwelcome economic effects (Bengui and Coulibaly, 2023)

CFMs can increase probability of binding credit constraints (Forbes, 2007, JIE)

**But, some countries still do this . . . why?**

Given these new circumstances and new data availability, we want to see:

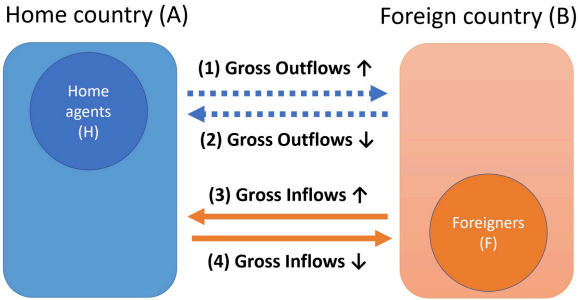
- Whether the CFMs have foreign shocks insulation effects on the net capital flows
- How does the effect vary on different dimensions? (flows, policies, assets.)

# Data

**Capital Controls**: We exploit recently available data that reports the application of CFM measures for 32 economies (***IMF Taxonomy of CFMs, 2019***) $\rightarrow$ CFMs and type (on inflows, outflows) reported.

**Net and Gross capital flows**: Computed with IMF-IFS data and along the lines of Forbes and Warnock (2012) and Cavallo, et al (2015, JIMF)

Inflows: Investments at by foreigners  Outflows: Investments abroad by domestic agents




**Gross Inflows** $= (3) - (4)$

**Gross Outflows** $= (1) - (2)$

**Net Inflows** $=$ Gross Inflows $-$ Gross Outflows

Note: We use a BOP perspective to report flows. Then: $\uparrow$ Inflows $\Rightarrow \uparrow$ Income ($\uparrow$ Outflows $\Rightarrow \downarrow$ Income).

# Data (cont.)

**Foreign Monetary Shock:** We include monetary policy surprises from Jarocinsky and Karadi (2020, AEJ Macro).

Based on Gertler and Karadi (2015) but refined to remove positive stock market effect: surprise on 3-month-ahead federal funds future rates around FOMC dates + SVAR based identification

*Sample:* 2000-2018 quarterly (emerging economies)

After adjusting for data limitations (and specification) our final panel consists of 13 economies

Note: the CFM policy data is still limited as it does not account for the intensity of the control measures.

# Empirical strategy and specification

Local Projection with AR distributed lags approach.

**Baseline:** Two-stage IV LP regression approach

CFMs: Hard to instrumentalize ⇒ Use lag to remove simultaneity bias.

Monetary Shock IV

$$
\begin{aligned}
y_{i,t+h} - y_{i,t-1} = \alpha^h + \beta_1^h \widehat{USMP}_t + \beta_2^h CFM_{i,t-1} + \beta_3^h (\widehat{USMP}_t \times CFM_{i,t-1}) \\
+\gamma^h IndividualControl_t + \delta^h GlobalControl_t \\
+\eta^h \sum_{j=1}^{J} Lag_{i,t-j} + \theta^h GFCdummy_t + FE_i^h + \epsilon_{i,t+h},
\end{aligned}
\tag{1}
$$

*First stage*: Estimate $\widehat{USMP}_t$

*Second stage*: Estimate $\beta_1^h, \beta_3^h$ (effects of shock on capital controls)

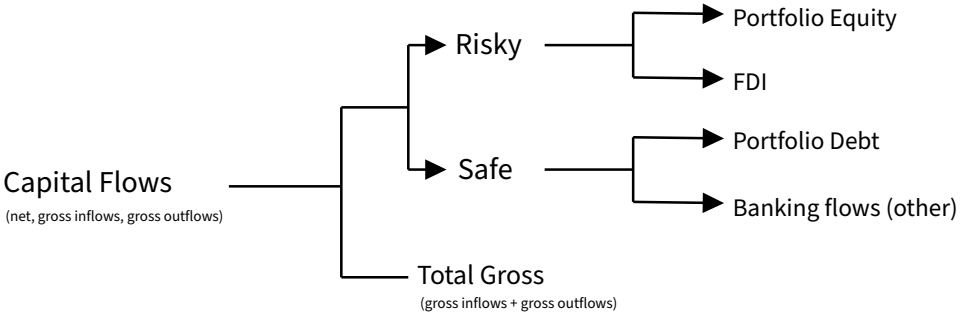
Controls (current and lags): (global) VIX, US growth rate; (individual) growth, inflation, ER depreciation, domestic interest rate

# Cases analyzed

**Quantity of interest:**

- effect on capital flows of (US MP) shock: $\beta_1^h$
- effect on flows of shock when CFMs are implemented: $\beta_1^h + \beta_3^h$

**Cases:** we analyze the effects in different types of flows and from different CFM policies:



# Results - Total Flows

**Blue:** ($\beta_1$) effect of foreign policy shock on capital flows

**Red:** ($\beta_1 + \beta_3$) effect of foreign shock on capital flows with CFMs

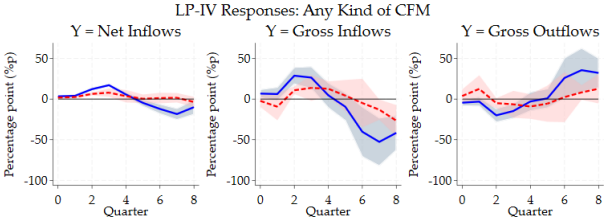



Figure: LP-IV IRFs to an Increase of 1% in the US MP Shock

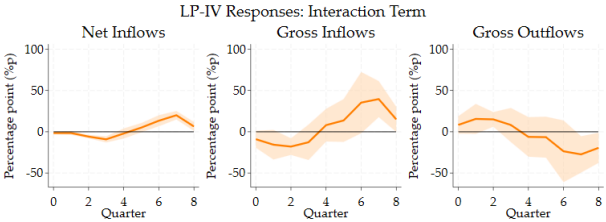
- **Mitigation**: effect is pushed towards origin for most horizons
- Net flows: Smaller effect $\rightarrow$ Interplay of stronger effects on Gross flows.
- Still, signs shift which seems odd $\rightarrow$ Better to disentangle effect (by assets, policies, regimes)

# Results (cont.)

Insulation effects on net flows: Not mild or ambiguous.

Instead, strong —offsetting— effects on the gross flows.

Reflected by the interaction term:

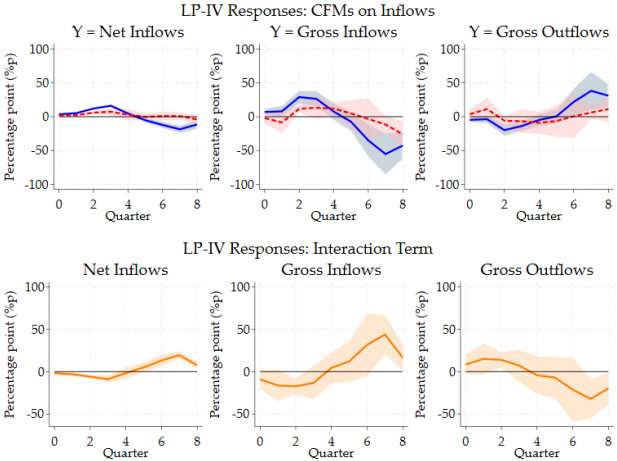



- Is the mitigation generated by specific types of CFM measures (on inflows, on outflows)?
- Can we recover net flows dynamics from the effects on inflows only? (as often thought) go

# Effects by type of CFM policy

Do policies targeting specific flows have different insulation features?

Feasible by analyzing only **CFMs on Inflows and on Outflows** (i.e., $CFM^{in}, CFM^{out}$, instead of $CFM$)




CFM on Inflows: Quite similar to the overall effects

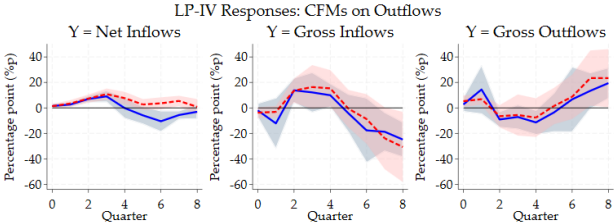
Mild mitigation for net flows: arises from strong effects on gross flows

Effects on Gross flows are stronger on inflows side

# Effects by type of CFM policy (cont.)

The effects on outflows are not as strong

On gross outflows the is virtually no mitigation (or is weaker)



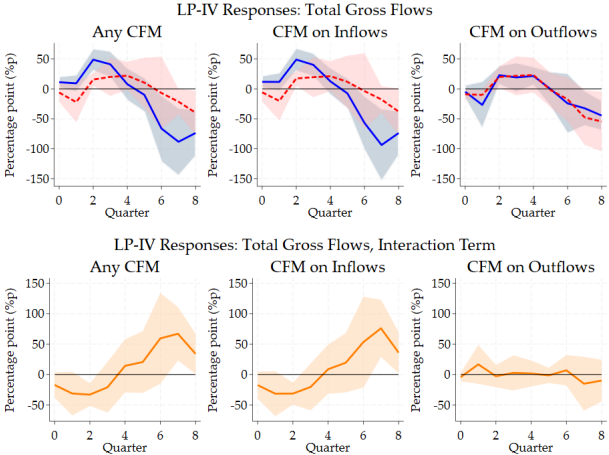

Effects seem **dictated by controls on inflows**

They can still affect the gross outflows dynamics

# Insulation effects in Total Capital Flows

Effect on total capital flows (Gross Inflows + Gross Outflows) → Proxy of Financial Integration

Davis, Valente and Van Wincoop (2021): Total flows comove strongly with a Global Financial Cycle




Scope for insulation

The insulation features show again

Effects similar as in separate gross flows but with different magnitudes

(As before) CFMs on inflows are the tools with most insulation features

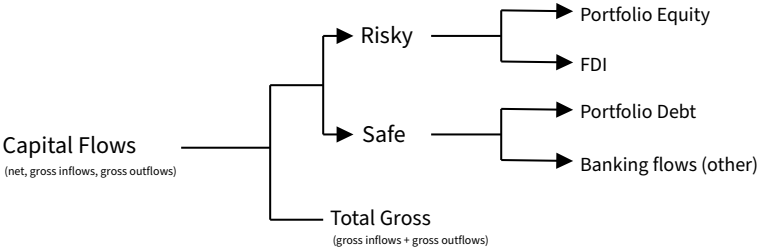
# Disaggregating Capital Flows

We can see there are insulation properties of the CFMs that help justify their implementation

However, the effects are not clear-cut (e.g., switching signs).

It is necessary to dig further:

- Risky vs. Safe assets: Is any of these protected more to external shocks by CFMs?
- Further subcategories:



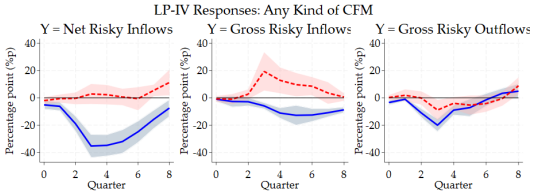

- Policy regimes

# Risk profile of flows

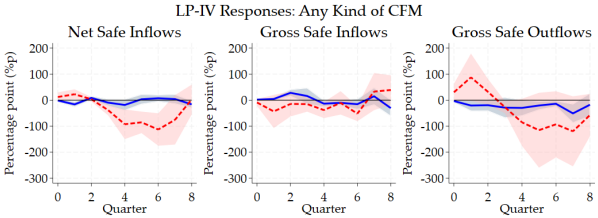
Insulation manifests for **Risky Flows only**

Useful for policy making as these are these are the "fickle" flows regulators want to stabilize.

(Farhi and Werning, 2014; Ghironi and Ozhan, 2021; Caballero and Simsek, 2020.)




Effects now show the expected signs.



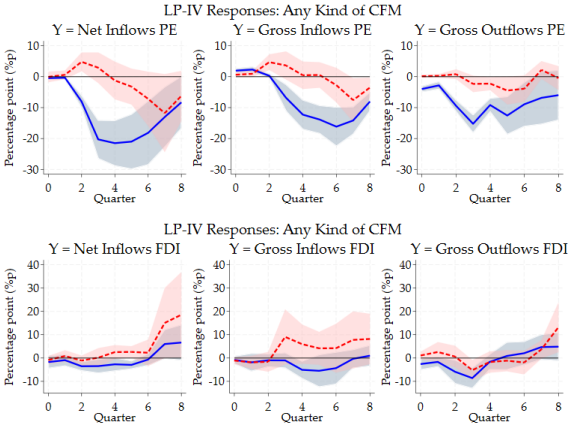

Now global retrenchment feature dominates compensation between gross flows.

Safe flows don't require insulation!

# Further disaggregation

We can explore further and see what, within the risky assets, depicts CFM insulation.

Mitigation effects are more strongly present for **portfolio equity** investments.




Further evidence that insulation focuses on fickle flows

FDI flows do not have capital flows effects (or insulation)

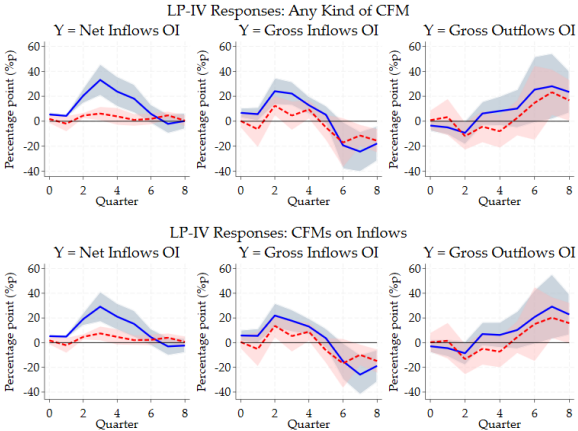
Insulation is stronger for Inflows controls too

Inflows CFMs - Risky

# Banking Flows

Even if "safe", Banking flows are known to elicit transmission of international shocks (Avjiev, Hardy, Kalemli-Ozcan, Serven, 2022)

In this case there is an effect to mitigate. However it's direction is contrary to the expected.




These flows explain the switching sings in the aggregate

Risk premium interpretation of the shock (higher foreign risk)

Higher implied return abroad becomes compatible with higher domestic inflows

# Effects by policy regime

Finally, the last dimension to explore is that of the shocks themselves.

Does the insulation change during tighter monetary conditions or by the business cycle pattern of the shocks?

Construct modified policy instruments for separate estimations along the lines of Jordà, Singh, and Taylor (2020) as follows:

$$z_t^{expansion} = \begin{cases} 0 & \text{if } z_t > 0 \text{ or } \Delta USMP_t > 0 \\ z_t & \text{otherwise} \end{cases}, \qquad z_t^{contraction} = \begin{cases} 0 & \text{if } z_t < 0 \text{ or } \Delta USMP_t < 0 \\ z_t & \text{otherwise} \end{cases}$$

The first type of shock implies the policy shock is always expansionary.

# Effects by policy regime

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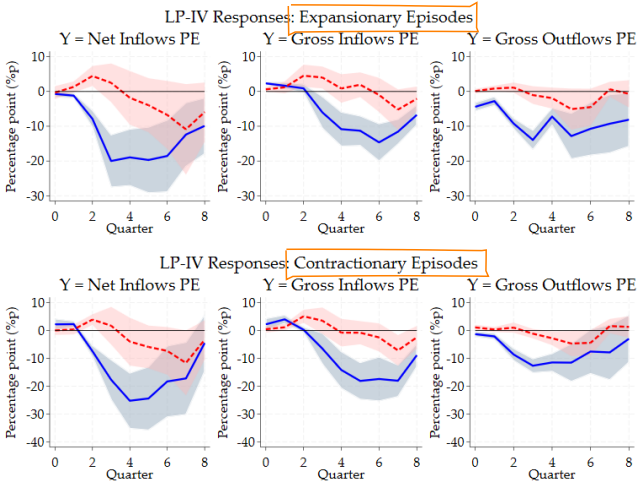


**Cycle-phase specific effects**:

Retrenchment stronger during tighter monetary conditions (or for contractionary shocks).

Insulation takes place in both regimes.

But is less effective during contractions —when needed most.

policy challenge

# Conclusions

- We tackle the question of whether the CFMs help insulate an economy from global shocks
- Relevant due to new environment: Capital dynamics in EMEs + Higher financial integration
- **Insulation occurs** at Net, Gross (In and Out) and Total Gross levels.

But the **level of disaggregation matters** (of flows, policies, and assets) for evaluating the controls:

- The mitigation takes place for **Risky Assets Flows and Banking Flows**.
  - Within Risky flows: Controls protect Portfolio Equity Flows only
  - Other flows (e.g., safe): Not as affected by the international shocks $\Rightarrow$ don't require insulation
- Thus: With too much aggregation $\longrightarrow$ bias in assessment of CFMs
  (overestimation: Since not all flows need insulation; underestimation: If only considering Net flows)
- Future research: Intensity of controls, and potential instrument construction for CFMs.

# Thank You!

Questions and feedback are welcome!

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# Robustness

*Domestic rates:* We lose a good amount of data by including the domestic rates. However it can be important.

The estimation without them still shows mitigation effects but it's harder to analyze as the dynamics can include domestic monetary policy responses to the shock

*GFC:* we control for the Global Financial Crisis episode with a dummy variable in the baseline, the estimation without it show similar effects.

*Lagged controls:* there is no golden rule as to how many lags of explanatory variables to add. We increase them and find similar results.

*Inclusion of other countries:* the only non-emerging country in the CFM database is Canada. When included, results are similar but depict a much higher volatility that prevent a clear interpretation.

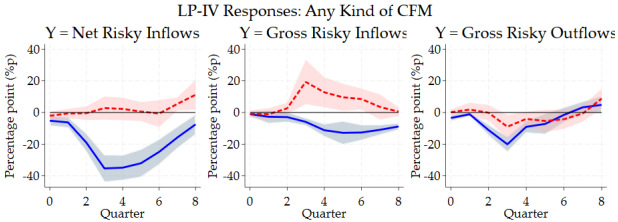
*Lagged CFM:* Mitigation effects without lagging the CFM measure variable, the effects are also similar.

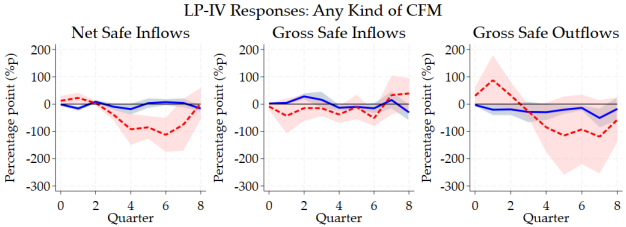
*Other:* no GDP scaling, $CFM_{t-2}$, more controls: Macroprudential policies, global rates, BRICS/Non-BRICS. back

# Risky flows and CFM on Inflows

Both policy tools generate insulation.

However it is still stronger on the Inflows' tool side.

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